Total Quality Management

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

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Total Quality Management (A09678)

Short Title:	Total Quality Management		
Faculty	Science and Computing		
Department	Science		
Cluster	Quality, Regulatory and Compliance		
Author	BWHELAN		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module covers aspects of quality management, including quality systems and standards, quality auditing, teamwork, human resource development and problem-solving techniques. It prepares the student for work in a pharmaceutical, food or agri-food company, which operate under strict guidelines for production and quality outputs.

Programmes

QUAL-0029	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	3 / 6 / M
QUAL-0029	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	3 / 5 / M
QUAL-0029	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	3 / 5 / M
QUAL-0029	BSc in Food Science with Business (WD_SFDSC_D)	3 / 6 / M
QUAL-0029	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	3 / 6 / M
QUAL-0029	BSc in Pharmaceutical Science (WD_SPHSC_D)	3 / 6 / M

Indicative Content

- Introduction to Quality Management: quality concepts, fitness for use, factors affecting quality
- Total Quality Management: quality improvement programmes, customer satisfaction
- Quality Auditing: types of audits, how to conduct a compliance audit
- Teamwork: leadership roles, working as part of a team, key requirements
- Human Resource Development: key aspects of training
- Problem Solving Techniques and their uses; Pareto Analysis, Cause and Effect Analysis, Brainstorming, 5 Why's, Statistical Process Control
- Overview of Quality System standards, documentation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Communicate in an organisation which has a quality system and standards.
2. Design a quality plan for auditing and compliance.
3. Examine professional behaviour and develop good communication skills.
4. Use techniques of critical thinking, problem-solving and decision making.
5. Interpret the various quality standards for preparation of relevant documentation.

Learning and Teaching Methods

- Lectures.
- Assignment.
- Self-directed learning.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4
Continuous Assessment	40%	
Assignment	40%	5

Assessment Criteria

<40%: No understanding of the basic concept of quality management. Unable to carry out critical thinking or problem solving techniques.

40%-49%: Is able to understand the quality management philosophies. May have some difficulty with critical thinking and problem solving.

50%-59%: All the above in addition to problem solving techniques and teamwork abilities. Correctly make suggestions for problem solving in an industrial environment.

60%-69%: In addition to the above, good critical thinking and decision making abilities. Shows a high level of competence and efficiency in the quality management techniques.

70%-100%: All previous to excellent level. Be able to describe and discuss quality management in a production environment and show initiative in approaching problems.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Essential Material

- "www.iso.org." http://www.iso.org/iso/home.html
- "www.nsai.ie." http://www.nsai.ie/Our-Services/Measurement/Calibration-Services.aspx
- James, P.T.J. _Total Quality Management_. New York: Prentice Hall, 1996.
- Kolarik, W.J. _Creating Quality - Concepts, Systems, Strategies and Tools_. London: McGraw-Hill, 1995.
- Lal, H. _Organisational Excellence through Total Quality Management: a practical approach_. New Delhi: New Age International, 2008.
- Shridhara Bhat, K. _Total Quality Management; test and cases_. India: Himalaya Publishing house, 2010.

Supplementary Material

- Taormina, T. _Successful internal auditing to ISO 9000_. London: Prentice Hall, 1999.
- Vorley, G. _Quality Management: princ__iples and practice_ . Guilford: Prentice Hall, 1998.